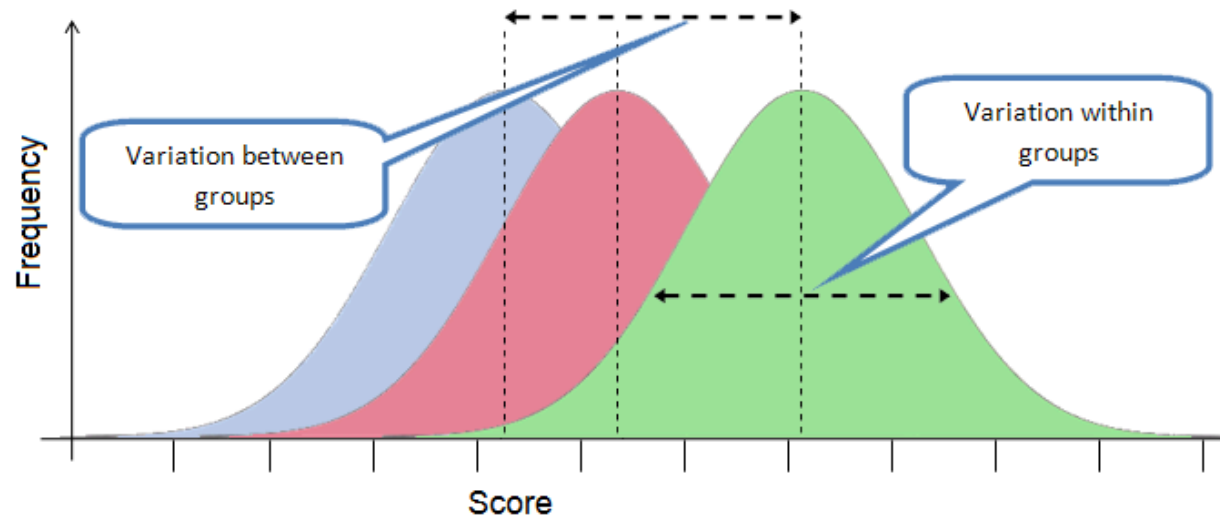


ANOVA: Analysis of Variance



It is a statistical method used to compare the means of **three or more independent groups**.

Instead of comparing every pair separately, ANOVA performs **one overall test** to determine whether at least one group mean differs from the others.

Suppose a teacher wants to compare the average marks of students taught using **three different teaching methods**.

Method	Average Marks
Method A	72
Method B	78
Method C	85

Question:

Which teaching method is better?

Can we use a t-test?

No.

A t-test compares **only two groups**.

Here, we have **three groups**.

This is where ANOVA is used.

Why Not Use Multiple t-tests?

Suppose we have four groups:

- A
- B
- C
- D

Pairwise comparisons:

- A vs B
- A vs C
- A vs D
- B vs C
- B vs D
- C vs D

That's **6 separate t-tests**.

Each test has a 5% chance of a Type I error.

Performing many tests increases the probability of obtaining a false positive.

ANOVA avoids this by using **one test**.

Real-Life Examples

Education

Compare three teaching methods.

Medicine

Compare four medicines.

Agriculture

Compare different fertilizers.

Marketing

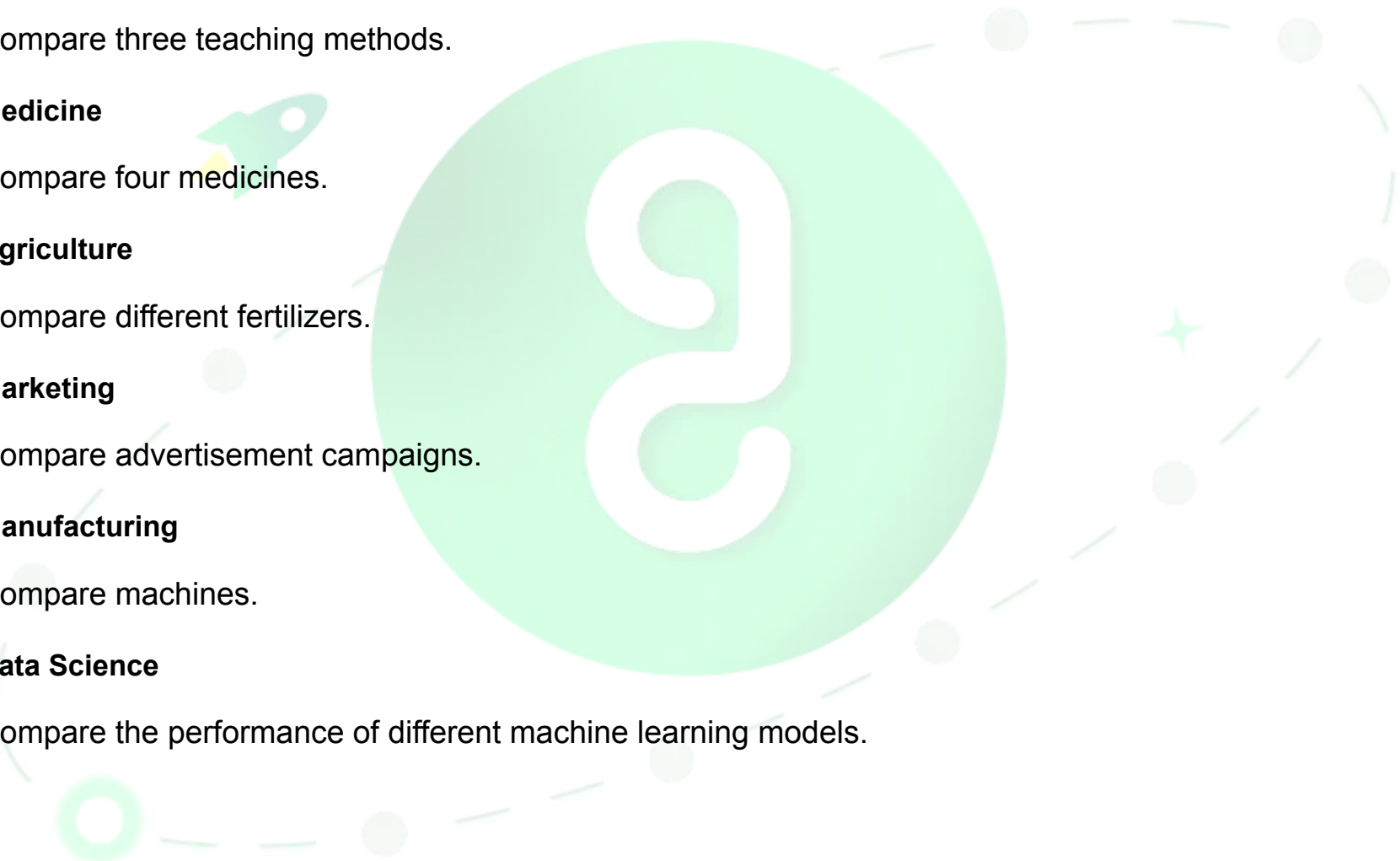
Compare advertisement campaigns.

Manufacturing

Compare machines.

Data Science

Compare the performance of different machine learning models.



Types of ANOVA

One-Way ANOVA

One independent variable.

Example:

Teaching Method → Student Marks

Two-Way ANOVA

Two independent variables.

Example:

Teaching Method + Gender → Marks

Repeated Measures ANOVA

Same participants measured multiple times.

Example:

Weight before, during, and after a diet program.

Hypotheses

Null Hypothesis (H_0)

All group means are equal.

$$H_0: \mu_1 = \mu_2 = \mu_3$$

Alternative Hypothesis (H_1)

At least one group mean differs.

Notice: ANOVA does **not** tell you **which** group is different.

It only indicates that **a difference exists**.

Assumptions of ANOVA

1. Random sampling
2. Independent observations
3. Normal distribution within each group
4. Equal variances (homogeneity of variance)

Understanding Variance

ANOVA compares **two sources** of variation:

Between-Group Variation

Variation caused by the different treatments or groups.

Example:

Average marks differ between teaching methods.

Within-Group Variation

Variation among individuals within the same group.

Example:

Students in the same class have different marks.

Key Idea

If the **between-group variation** is much larger than the **within-group variation**, the groups are likely to have different means.

ANOVA Workflow

1. Define the problem.
2. State H_0 and H_1 .
3. Collect sample data.
4. Calculate group means.
5. Calculate the grand mean.
6. Calculate SST.
7. Calculate SSB.
8. Calculate SSW.
9. Calculate Mean Squares.
10. Calculate the F-statistic.
11. Find the p-value.
12. Make a decision.

Components of ANOVA

Total Sum of Squares (SST)

Measures the **total variation** in the data.

Between Sum of Squares (SSB)

Measures variation due to differences between group means.

Within Sum of Squares (SSW)

Measures variation within each group.

Relationship

$$SST = SSB + SSW$$

This is one of the most important concepts in ANOVA.

Degrees of Freedom

- Between groups:
df $B = k - 1$
where k is the number of groups.
- Within groups:
df $W = N - k$
where N is the total number of observations.
- Total:
df $T = N - 1$

Mean Squares

Mean Square Between

$$MSB = \frac{SSB}{df\ B}$$

Mean Square Within

$$MSW = \frac{SSW}{df\ W}$$

F Statistic

$$F = \frac{MSB}{MSW}$$

Interpretation:

- Large F: Groups differ significantly.
- Small F: Differences are likely due to random variation.

ANOVA Table

Source	SS	df	MS	F
Between Groups	SSB	k-1	MSB	F
Within Groups	SSW	N-k	MSW	
Total	SST	N-1		

Decision Rule

- If $p < 0.05$, reject H_0 .
- If $p \geq 0.05$, fail to reject H_0 .

ANOVA vs t-test

Feature	t-test	ANOVA
Groups Compared	2	3 or more
Test Statistic	t	F
Purpose	Compare two means	Compare multiple means
Output	Difference between two groups	Whether at least one group differs

ANOVA Example

Problem Statement

A school wants to determine whether three different teaching methods produce different average marks.

Data

Method A	Method B	Method C
10	15	20
12	17	22
14	19	24

Question:

Are the average marks significantly different?

Step 1: State the Hypotheses

Null Hypothesis

The average marks are equal.

$H_0: \mu_A = \mu_B = \mu_C$

Alternative Hypothesis

At least one average is different.

H_1 : At least one mean differs

Step 2: Calculate Group Means

Method A

$$\frac{10+12+14}{3} = 12$$

Method B

$$\frac{15+17+19}{3} = 17$$

Method C

$$\frac{20+22+24}{3} = 22$$

Group Means

Group	Mean
A	12
B	17
C	22

Step 3: Calculate Grand Mean

First find total.

$$10+12+14+15+17+19+20+22+24 = 153$$

Total observations

$$N=9$$

Grand Mean

$$\bar{x} = \frac{153}{9} = 17$$

Step 4: Calculate SSB (Between Group Sum of Squares)

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

Where

- $n_i = 3$
- Grand Mean=17

Group A	12-17	25	25x3 = 75
Group B	17-17	0	0x3 = 0
Group C	22-17	25	25x3 = 75

Therefore

$$SSB = 75 + 0 + 75 = 150$$

Step 5: Calculate SSW (Within Group Sum of Squares)

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

This measures the variation within each group.

Group A

Mean = 12

Value	Difference	Square
10	-2	4
12	0	0
14	2	4

Total

$$4+0+4=8$$

Group B

Mean=17

Value	Difference	Square
15	-2	4
17	0	0
19	2	4

Total : 8

Group C

Mean=22

Value	Difference	Square
20	-2	4
22	0	0
24	2	4

Total : 8

Therefore

$$\mathbf{SSW} = \mathbf{SSW}_A + \mathbf{SSW}_B + \mathbf{SSW}_C = 8 + 8 + 8 = 24$$

Step 6: Calculate SST

Formula

$$\mathbf{SST} = \mathbf{SSB} + \mathbf{SSW}$$

$$= 150 + 24$$

$$= 174$$

$$\mathbf{SST} = 174$$

Step 7: Degrees of Freedom

Number of groups : $\mathbf{k} = 3$

Total observations : $\mathbf{N} = 9$

Between - $\mathbf{df\ B} = \mathbf{k} - 1 = 3 - 1 = 2$

Within $\mathbf{df\ W} = \mathbf{N} - \mathbf{k} = 9 - 3 = 6$

Total $\mathbf{df\ T} = \mathbf{N} - 1 = 8$

Step 8: Mean Squares

Between

$$MSB = \frac{SSB}{dfB} = \frac{150}{2} = 75$$

Within

$$MSW = \frac{SSW}{dfW} = \frac{24}{6} = 4$$

Step 9: Calculate F Statistic

$$F = \frac{MSB}{MSW} = \frac{75}{4} = 18.75$$

Step 10: Create ANOVA Table

Source	SS	df	MS	F
Between Groups	150	2	75	18.75
Within Groups	24	6	4	
Total	174	8		

Step 11: Calculate P-Value with F Statistic

For

- $\alpha = 0.05$
- F Statistic = 18.75
- $df_n = df_B = 2$
- $df_d = df_W = 6$

As P-Value [Calculator](#),

P - Value = 0.0026

Since: **0.0026 < 0.05**

Reject the null hypothesis.

Step 12: Conclusion

There is sufficient statistical evidence to conclude that **at least one teaching method has a significantly different average score**.

Note: ANOVA does **not** tell us which teaching methods differ. A **post hoc test** (e.g., Tukey's HSD) is required for pairwise comparisons.

Final Summary

Quantity	Between Group	Within Group	Total	General
Mean A				15
Mean B				17
Mean C				22
Grand Mean				17
Sum of Square	150	24	174	
df Between	2	6	8	
Mean Square	75	4		
F				18.75
P value				0.0026
Decision				Reject H_0